

Project Close-out Report for: Rapid DEX: Mainnet-Ready, Batcher-Free & Advanced Features

Name of Project and Project URL on Catalyst / IdeaScale

- **Name:** Rapid DEX: Mainnet-Ready, Batcher-Free & Advanced Features
 - **Project URL:** [Project Catalyst Fund 14](#)
 - **Project ID:** 1400104
 - **Project Category:** Cardano Open Developers
 - **Name of Project Manager:** Roman Majovsky
 - **Date Project Started:** Nov 24, 2025
 - **Date Project Completed:** April 30, 2026
-

List of Challenge KPIs and How the Project Addressed Them

1. Mainnet Readiness for a Batcher-Free DEX

The project focused on advancing Rapid DEX from an earlier proof of concept into a mainnet-ready architecture by delivering production-level smart contracts, modular infrastructure, and end-to-end application workflows.

2. Improved User Experience for Liquidity Management

Traditional liquidity management often requires multiple steps and balanced asset deposits. Rapid DEX introduced Zap In and Zap Out functionality, enabling one-token entry and exit flows that significantly simplify user interaction.

3. Sustainable Pool Economics

The project implemented flexible swap fees and built-in treasury support, allowing pool creators to define custom fee logic, treasury participation, and sustainable monetization models.

4. Open Infrastructure for Ecosystem Builders

By publishing SDK packages, API documentation, and modular repositories, the project reduced integration barriers for wallets, aggregators, and third-party developers building on Cardano.

List of Project KPIs and How the Project Addressed Them

1. Milestone 1 – Smart Contract Delivery

Delivered open-source pool smart contracts supporting:

- Pool creation

- Token swaps
- Liquidity deposits and withdrawals
- Treasury withdrawals
- Donations to pools
- Flexible fee configuration
- Zap In / Zap Out functionality

All contracts compile successfully, passed unit tests, and were published with documentation and build instructions.

Relevant sources:

- [Smart Contracts Repository](#)
- [Validator Contract](#)
- [Build Instructions](#)

2. Milestone 2 – Developer SDK & Modular Architecture

Released modular SDK packages for ecosystem integrations:

- [@wingriders/rapid-dex-sdk-core](#)
- [@wingriders/rapid-dex-sdk-react](#)

The repository architecture was reorganized into clearly separated modules, improving maintainability, scalability, and future extensibility.

Relevant sources:

- [SDK Source Code](#)
- [Architecture Documentation](#)

3. Milestone 3 – Backend & Frontend Application

Delivered a working public staging application with:

- Wallet connection
- Pool creation
- Swaps
- Liquidity management
- Treasury operations
- Real-time updates

The application was successfully deployed to mainnet, enabling live user interactions.

Relevant sources:

- [Rapid DEX App](#)
- [Main Repository](#)
- [API Documentation](#)

4. Transparency & Documentation KPI

Published architecture documentation, API references, test evidence, walkthrough videos, and open repositories for full public review.

Relevant sources:

- [Architecture Docs](#)
-

Key Achievements

- Successfully completed all planned Catalyst milestones
 - Deployed Rapid DEX to mainnet
 - Built production-grade smart contracts with advanced pool logic
 - Released reusable SDK packages for ecosystem integrations
 - Demonstrated complete user workflows in a public staging environment
 - Enabled configurable pool fees and treasury mechanics
 - Simplified LP onboarding through Zap In / Zap Out flows
 - Maintained fully open-source development and public transparency
-

Key Learnings

1. Batch-Free Architecture Can Scale Further

Direct pool interactions remain a strong model for efficient and decentralized trading infrastructure.

2. UX Innovation Drives Adoption

Features such as Zap In / Zap Out and configurable fee models improve accessibility for a wider range of users.

3. Modular Tooling Accelerates Ecosystem Growth

Reusable SDKs and documented APIs lower integration costs and help external teams build faster.

4. Real-Time Feedback Builds Confidence

Live updates, visible transaction progress, and responsive interfaces improve trust in on-chain products.

Ecosystem Value

Rapid DEX contributes reusable open-source infrastructure that can be integrated, extended, or studied by the wider Cardano ecosystem.

The project supports:

- Wallet integrations
 - Aggregator integrations
 - New liquidity models
 - Developer experimentation
 - More competitive decentralized exchange infrastructure on Cardano
-

Next Steps for the Product or Service Developed

Potential next priorities include:

- Independent smart contract security audit
 - Additional wallet integrations
 - Aggregator partnerships
 - Expanded analytics dashboards
 - Additional routing and liquidity features
 - Performance benchmarking under higher transaction load
 - Broader ecosystem integrations
-

Final Thoughts / Comments

Rapid DEX: Mainnet-Ready, Batchers-Free & Advanced Features successfully transformed an earlier concept into a significantly more mature decentralized trading platform.

The project delivered advanced smart contract functionality, developer tooling, and a live application environment already deployed in production.

We thank the Catalyst community for supporting this proposal and look forward to the next phase of Rapid DEX development on Cardano.

Links to Relevant Project Sources

- Application: [Rapid DEX App](#)
- Main Repository: [GitHub – rapid-dex](#)
- Smart Contracts: [GitHub – rapid-dex-contracts](#)

SDK Packages

- [@wingriders/rapid-dex-sdk-core](#)
- [@wingriders/rapid-dex-sdk-react](#)

Documentation

- [Architecture Docs](#)
- [API Documentation](#)
- [Build Instructions](#)

Evidence & Testing

- [Compilation & Tests Recording](#)
- [Milestone 3 Demo Video](#)
- [Transaction Chaining Test](#)

Close-out Deliverables

- [Close-out Video](#)